

Part 2:

Download the preliminary website; it has the set of requirements outlined in the Home page, on the Technology Topic page and Guidelines pages. You may use this preliminary website as the basis for your website, or you may design an entirely new website. Your group website should address the requirements outlined in the preliminary website and this document.

As working with the group progresses, keep track of the group process (See Guidelines Page of preliminary website) using e.g., Meeting Minutes and any other files evidencing your group's work.

Only one group member needs to submit on behalf of the group. The group needs to agree on who

will submit on behalf of the group and document this in their meeting minutes. Group members also

need to agree on a method to confirm that upload has been done (E.g., This may be via a message

from the person confirming to group members that submission was done "on this date, at this time"). This is so that your work can be marked.

Requirements for your website include:

❏ Scholarly expectations of design and referencing.

❏ You should refer mostly to academic articles (at least 10); however, reference to websites in addition to this is acceptable. (cited as APA formatted references)

❏ Details of the group process as indicated in the Guidelines Page including: Meeting Minutes and files evidencing Groupwork.

❏ Must have at least 1-3 figures (or images) on website

Work Allocation:

Group members: Matthew & Dylan

Section	Matthew	Dylan
Section 1 — Technology / Topic	✓ 21/04/2026	Edited 29/04/2026
Section 2 — Opportunities	✓ 07/05/2026	
Section 3 — Risks		✓ 03/05/2026
Section 4 — Choices		✓ 19/05/2026
Section 5 — References	✓	✓
Website	✓ Edited/Updated 29/05/2026	✓ Created 22/05/2026

1. Technology / Topic

To understand why systems like Starlink represent a meaningful shift in satellite communications, it helps to start with the physics. Satellites don't all orbit at the same altitude where a satellite sits in space has a profound effect on how useful it is for communications, and the differences between orbital classes are not subtle.

Traditional communication satellites operate in what's called geostationary orbit, or GEO. At roughly 36,000 kilometres above the Earth's surface, a satellite in GEO moves at exactly the same angular speed as the Earth's rotation, meaning it appears to hover in a fixed position over one spot on the equator. This is genuinely useful: a single GEO satellite can see about a third of the planet's surface, and a dish pointed at it never has to track a moving target. For broadcast television and older satellite internet services, this made GEO the dominant choice for decades.

The problem is the distance. At 36,000 km, a signal travelling at the speed of light still takes roughly 240 milliseconds just to make the one-way trip from ground to satellite. Round-trip latency the time it takes to send a request and get a response sits around 600 milliseconds or more for GEO systems. That's acceptable for watching a satellite TV broadcast, but it makes real-time applications like video calls, online gaming, or cloud-based work frustrating or unusable. It also creates headaches for protocols like TCP/IP that were designed assuming much shorter round-trip times.

Low Earth Orbit, by contrast, places satellites at altitudes typically ranging from around 400 to 1,200 kilometres. Starlink's operational shells sit mostly around 550 km. At that distance, the signal travel time drops dramatically round-trip latency for Starlink users has been measured at roughly 25 to 60 milliseconds under good conditions, which is broadly comparable to a ground-based broadband connection. This is the central technical advantage that makes LEO satellite broadband worth taking seriously.

However, proximity to Earth introduces an immediate complication: coverage. A LEO satellite at 550 km sees only a comparatively small footprint on the Earth's surface — and, critically, it's moving fast. A single LEO satellite completes a full orbit in roughly 90 to 95 minutes, streaking across the sky in a matter of minutes from the perspective of a ground observer. For continuous coverage of any given point on Earth, you need hundreds or thousands of satellites arranged across multiple orbital planes, timed so that there's always at least one overhead with a clear line of sight. This is the logic behind the constellation model — Starlink had deployed over 7,000 satellites by end of 2025.

The word "mesh" in this context refers to something specific and technically important. In networking, a mesh topology is one where nodes maintain direct connections to multiple neighbours, and data can hop node-to-node along multiple paths rather than always returning to a central hub. Early Starlink used a "bent-pipe" relay model: your dish sends a signal up to a satellite, which reflects it down to a nearby ground station, which then connects to the internet. The satellite itself does minimal processing. This isn't really a mesh — it requires a ground station within the satellite's coverage footprint, which limits usefulness over oceans and remote regions.

The shift to a genuine mesh comes from inter-satellite links, or ISLs — direct laser communication links between satellites themselves. Newer Starlink satellites use laser-based ISLs to pass data laterally across the constellation in space before it descends to a ground station. Laser links carry very high bandwidth over long distances with minimal interference, since the absence of atmosphere eliminates the scattering that limits ground-based laser communication. With ISLs in place, a user in a remote area can send data up to a satellite, have it routed through multiple ISL hops across the constellation, and descend to a ground station thousands of kilometres away — without any ground infrastructure beneath most of that path.

The routing challenge this creates is genuinely complex. Satellites orbit at roughly 7.5 km/s, so the network topology — which satellites can see each other — changes continuously. Routing algorithms must predict future link states and pre-plan paths that remain valid long enough to be useful. Research has produced approaches like time-expanded graphs, where known future satellite positions are used to compute stable routes across handoff sequences, and on-demand schemes that recalculate dynamically. Managing this distributed, time-varying network at scale, with thousands of moving nodes and millisecond-level routing decisions, is one of the genuinely hard engineering problems in modern networking.

2. Opportunities

The most significant opportunity presented by LEO satellite broadband is its potential to reach people and places that conventional infrastructure has consistently failed to serve. The "digital divide" — the gap between those with reliable internet access and those without — maps closely onto geography. Dense urban areas attract fibre and mobile investment because population density makes the economics work; rural and remote areas often don't clear that threshold. The residents of those areas have disproportionately borne the costs in terms of education, healthcare, economic opportunity, and social connection.

The policy and academic literature is broadly consistent on why this matters. Access to broadband shapes students' ability to participate in remote learning and do schoolwork requiring online resources. It affects small businesses' ability to reach markets and access

services that increasingly exist only online. It determines the quality of healthcare available in rural areas, where telehealth can substitute for long journeys to reach specialists. It influences whether young people in remote communities feel connected to broader social and cultural life. Research consistently finds meaningful correlations between broadband access and economic development outcomes, educational attainment, and community resilience — these are not marginal effects.

Against this backdrop, LEO satellite broadband represents a genuinely different proposition than previous satellite internet. The latency reduction relative to GEO — from over 600 milliseconds to the 25–60 millisecond range — is not just a technical nicety. Studies have measured Starlink latency averaging around 28 milliseconds compared to 685 milliseconds for comparable GEO services in rural deployments. That gap makes video calls, cloud software, and modern web applications actually usable in a way they simply weren't for earlier satellite subscribers. For a remote community in rural New Zealand, a farming region in sub-Saharan Africa, or a fishing village in the Pacific Islands, this is the difference between a service that works for contemporary needs and one that doesn't.

The coverage argument is also real. Where laying fibre is too expensive or slow, and where mobile towers don't reach, a constellation serving from orbit can provide connectivity regardless of terrain. The technology works equally well — in principle — over mountains, deserts, oceans, and polar regions that ground infrastructure struggles to reach. For maritime users, aviation, and emergency response in disaster-affected areas where terrestrial infrastructure has been damaged, this flexibility has concrete value.

There is also a resilience dimension. A distributed constellation with no single point of failure is harder to knock out than a centralised ground-based system. For critical services — emergency communications, remote healthcare coordination, disaster response — the redundancy of a mesh network spread across thousands of satellites offers genuine robustness advantages over systems that depend on a specific piece of ground infrastructure remaining intact.

The most credible framing in the literature positions LEO satellite broadband as a complement to fibre and mobile networks rather than a replacement — filling gaps those technologies leave rather than substituting for them wholesale. Fibre remains superior in throughput, latency consistency, and cost per bit at scale. Mobile broadband is often more affordable and serves use cases satellite doesn't. The realistic picture is a tiered approach: fibre and mobile where the economics work, satellite for the long tail of locations where they don't. Within that framing, the opportunity is real and substantial.

3. Risks

Any honest account of LEO satellite broadband has to engage with genuine risks — some specific to current systems, others raising longer-term concerns about sustainability, fairness, and unintended consequences.

****Variable and unreliable performance** is the most immediate practical issue. Unlike fibre, Starlink performance fluctuates significantly depending on conditions. Weather degrades signal strength; heavy rain or dense cloud cover can cause outages. Terrain matters — the dish requires a clear sky view, and locations with nearby trees, buildings, or steep hillsides experience regular interruptions as satellites pass behind obstructions. Time of day and local user density affect available capacity. Measurement studies consistently find a spread of performance outcomes rather than a single reliable figure. While median performance has improved as the constellation has grown, this variability is a real limitation for applications requiring consistent, predictable connectivity — and there is a meaningful risk of overselling the technology to underserved communities and then failing to deliver on those promises.

Orbital debris and long-term space sustainability are arguably the most serious risks. Each Starlink satellite has an operational lifespan of roughly four to six years, after which it deorbits. With over 7,000 satellites currently operational and plans for tens of thousands more from Starlink and competing constellations, low Earth orbit is becoming significantly more congested. The danger is not merely the presence of functioning satellites, but the potential for collisions. A collision between two objects in LEO generates a cloud of debris fragments travelling at orbital velocity, each capable of damaging or destroying other satellites. In the scenario described by researcher Donald Kessler — known as Kessler Syndrome — a cascading chain of collisions could render certain orbital altitudes unusable for generations. While this remains a tail risk rather than an imminent outcome, the decisions being made now about constellation scale, satellite design, and end-of-life disposal will shape the orbital environment for decades. This is a case where the costs of getting it wrong are large, hard to reverse, and borne by everyone — including future users who had no say in the decisions.

Impacts on astronomy are a more immediate and less speculative harm. The bright streaks left by Starlink satellites across long-exposure astronomical images have been well documented and widely criticised by the scientific community. Ground-based telescopes — including those used to detect near-Earth asteroids and conduct wide-field sky surveys — have had observations degraded or contaminated by satellite trails. SpaceX has made efforts to reduce satellite brightness through design changes, with partial success, but the fundamental issue of thousands of fast-moving objects crossing the sky persists. This is a case where the commercial interests of one organisation create a negative externality for a separate community — scientific researchers — who had no meaningful voice in the decision to launch at scale.

Cost and accessibility barriers** undercut the equity case. Hardware costs for a Starlink terminal and ongoing monthly subscription fees remain significant in precisely the communities where the access gap is most acute. A service that is technically available from orbit but

practically unaffordable for the intended beneficiaries doesn't fully deliver on its promise. Without deliberate policy intervention — subsidy programmes, community access models, digital literacy support — the gap between theoretical and actual benefit can be substantial.

Concentration of control is a less-discussed but important systemic risk. A small number of private companies — primarily SpaceX — currently dominate LEO satellite broadband. The infrastructure these companies are building is becoming critical for communities with no alternative. Questions of pricing, service terms, data sovereignty, and continuity of service are therefore significant, and the leverage available to affected communities or governments is limited when alternatives are absent.

4. Choices

The risks outlined above are not simply features of the technology they are the result of choices. Understanding who makes those choices, what alternatives exist, and what frameworks might guide better decisions is the core of any serious engagement with the ethics of LEO satellite broadband.

SpaceX chose to launch Starlink at scale rapidly, before comprehensive international frameworks for managing orbital congestion were in place. This was a commercially rational decision — moving quickly secured spectrum rights, established market presence, and created network effects that are difficult for later entrants to overcome. But it was a choice, not an inevitability, and it created facts on the ground — or rather, in orbit — that now constrain what other stakeholders can do. Governments and international bodies largely allowed this to happen, either because regulatory frameworks didn't anticipate the scale of commercial LEO deployment or because the economic and strategic interests involved made aggressive regulation unattractive.

For engineers and technical professionals working on these systems, professional codes of ethics are directly relevant. The ACM Code of Ethics calls on computing professionals to consider the public good, to be honest and trustworthy, and to avoid harm — including hard-to-reverse changes to shared environments. These aren't abstract principles: they create real obligations to raise concerns about long-term orbital sustainability, to push back when short-term commercial pressure overrides longer-term thinking, and to be honest with policymakers and the public about what the technology can and cannot deliver. The choice to remain silent when aware of significant systemic risks is itself a choice.

At the regulatory level, there are meaningful decisions still to be made. International coordination on debris mitigation standards — including requirements for satellite lifespan, deorbit timelines, and collision avoidance capabilities — is technically achievable but requires political will to implement. The ITU (International Telecommunication Union) governs spectrum allocation and has some role in orbital slot coordination, but its frameworks were not designed for constellations of thousands of satellites and are under pressure to adapt. The choice of whether to strengthen these frameworks, and how quickly, will shape the sustainability of the orbital environment.

At the policy level, the choice of how to deploy LEO broadband for public benefit rather than purely private profit involves subsidy design, procurement policy, and community engagement. Countries that have simply left deployment to the market have seen uneven outcomes — the economics of satellite broadband still tend to favour users who can pay, and communities most in need of access are not always the most commercially attractive to serve. Policy choices around who gets subsidies, how community access hubs are funded, and what data sovereignty requirements apply to satellite services used by public institutions all shape whether the technology's equity potential is realised.

For end users and communities, the relevant choices include whether to advocate for better regulation, how to engage with service providers about pricing and terms, and how to frame the technology in local infrastructure planning. Framing LEO broadband as a complement to rather than a substitute for fibre investment is a choice with significant long-term implications — communities that accept satellite service as sufficient may find their case for fibre infrastructure weakened, even though the long-term trajectory of fibre performance and cost is more favourable.

The tension at the heart of this technology — between genuine social benefit and real systemic risks — does not resolve itself. It requires ongoing choices by engineers, policymakers, companies, and communities about what kind of future they are building toward.

5. References

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